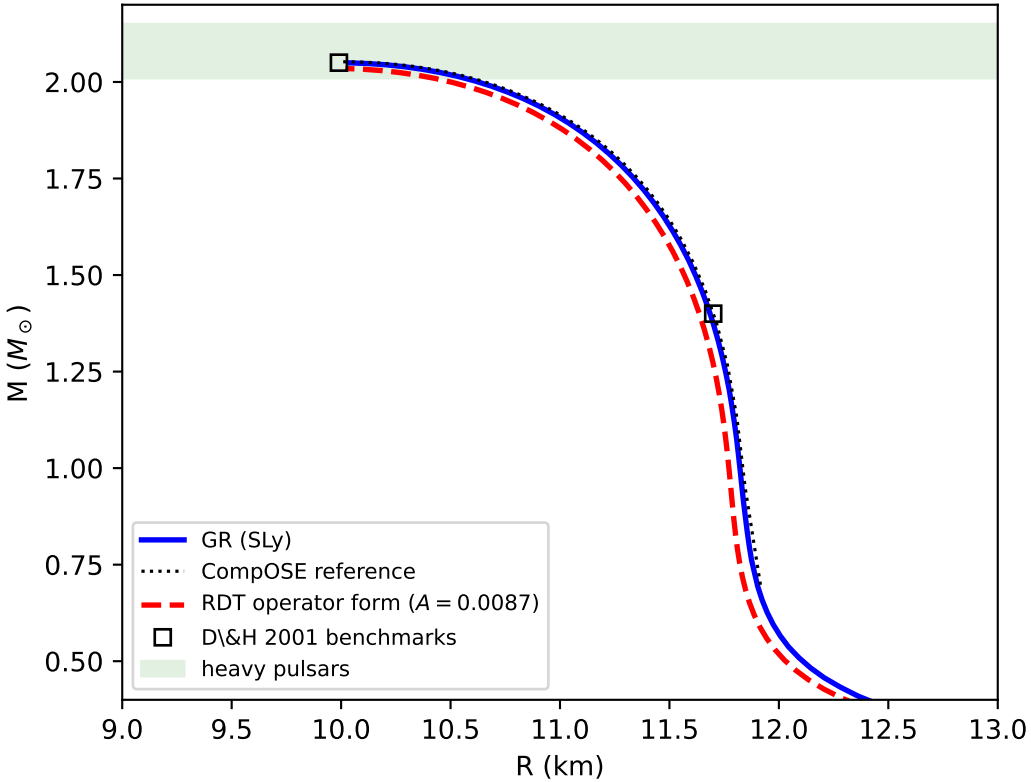
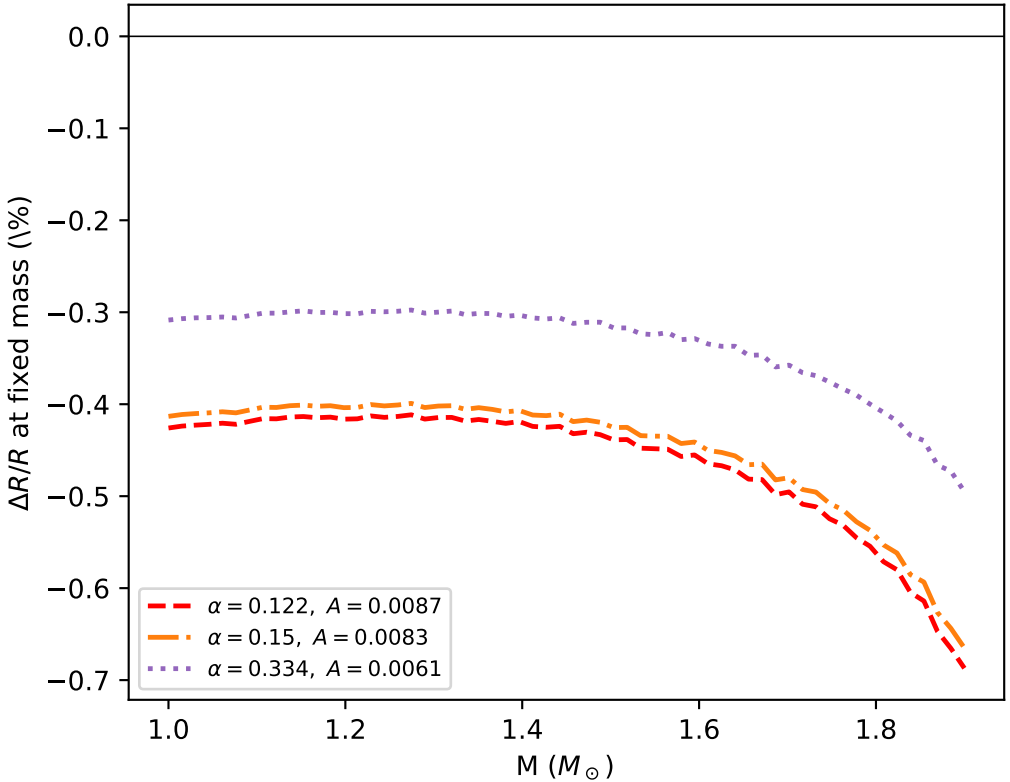

(a) Mass--radius: GR vs solar-allowed RDT (operator form)
M ($M_\odot$)
R (km)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
9.0
9.5
10.0
10.5
11.0
11.5
12.0
12.5
13.0
GR (SLy)
CompOSE reference
RDT operator form ($A = 0.0087$)
D\&H 2001 benchmarks
heavy pulsars


(b) Radius contraction along the solar exclusion boundary
$\Delta R/R$ at fixed mass (\%)
M ($M_\odot$)
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
−0.7
1.0
1.2
1.4
1.6
1.8
$\alpha = 0.122,\ A = 0.0087$
$\alpha = 0.15,\ A = 0.0083$
$\alpha = 0.334,\ A = 0.0061$